

Topological Dissipative Data Injection Theorem (TDDIT): Overcoming Quantum Input-Output Bottlenecks via Steady-State Invariants

Muhammad Aidil Amry
Independent Researcher
South Sulawesi, Indonesia
aidilamrym@gmail.com
ORCID: 0009-0002-9718-9710

September 2026

Abstract

In Quantum Data Science and Quantum Informatics, the fundamental barrier to practical implementation is the “Input-Output Bottleneck” associated with Quantum Random Access Memory (QRAM) limitations. Loading a classical dataset $X \in \mathbb{R}^{N \times d}$ into a quantum state $|\psi\rangle$ requires exponential time or space $\mathcal{O}(2^n)$, effectively destroying any quantum computational advantage. Furthermore, Parameterized Quantum Circuits (PQC) for Quantum Machine Learning (QML) inevitably suffer from Barren Plateaus, where gradients vanish exponentially with circuit depth. Modern science has failed to discover high-dimensional data mappings that do not destroy entanglement or become intractable within unmeasurable Hilbert spaces. This paper introduces the Topological Dissipative Data Injection Theorem (TDDIT), a globally novel mathematical framework that bypasses these limitations by mapping classical datasets into the Lindblad jump operators of an open quantum system, embedding high-dimensional geometric structures into macroscopic topological invariants.

1 The Global Information Gap

The current trajectory of Quantum Machine Learning (QML) relies heavily on unitary state preparation routines, such as $U(X)|0\rangle^{\otimes n}$. This paradigm is fundamentally flawed. State preparation for arbitrary data requires $\mathcal{O}(2^n)$ gates, and subsequent variational optimization faces Barren Plateaus ($\text{Var}(\partial_\theta \langle H \rangle) \in \mathcal{O}(2^{-n})$). The epistemological blind spot lies in the assumption that classical data must be mapped to the state vector directly via Hamiltonian evolution or unitary gates.

2 Mathematical Formulation: Topological Dissipative Data Injection

Instead of employing unitary operations, the classical dataset is mapped into the Lindblad jump operators of an Open Quantum System. Let $X \in \mathbb{R}^{N \times d}$ be a high-dimensional data matrix. We define a data-dependent Lindbladian Superoperator $\mathcal{L}_X$:

$$\frac{\partial \rho(t)}{\partial t} = \mathcal{L}_X[\rho(t)] = -i[H_0, \rho(t)] + \sum_{j=1}^d \gamma_j(X) \left(L_j \rho(t) L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho(t)\} \right) \quad (1)$$

Here, $\gamma_j(X)$ represents the dissipation rates mapping the Principal Components of the dataset X , and L_j are local quantum operators.

The system asymptotically evolves towards a unique steady-state density matrix $\rho_{ss}(X)$ such that:

$$\mathcal{L}_X[\rho_{ss}(X)] = 0 \quad (2)$$

Information extraction is completely detached from exponential state tomography. Instead, it relies on measuring the topological invariant of ρ_{ss} , specifically the Thermal Chern Number $\mathcal{C}(X)$:

$$\mathcal{C}(X) = \frac{i}{2\pi} \text{Tr} \left(\rho_{ss}(X) \left[\frac{\partial \rho_{ss}}{\partial x_\mu}, \frac{\partial \rho_{ss}}{\partial x_\nu} \right] \right) \quad (3)$$

This invariant encodes the macroscopic geometric structure of the classical dataset X and remains immune to local decoherence.

3 Formal Verification and Red-Teaming (SMT/Logic Protocol)

3.1 Iteration 1: Standard Unitary Embedding Testing

Hypothesis: Map matrix X to $U(x)|0\rangle$.

Defect Found (UNSAT): Gradient variance $\text{Var}(\partial_\theta \langle H \rangle) \in \mathcal{O}(2^{-n})$. The Barren Plateau is reached. Memory complexity is $\mathcal{O}(2^n)$.

Correction: Discard unitary evolution. Utilize dissipative evolution towards a steady-state.

3.2 Iteration 2: Dissipative Lindblad Evolution $\mathcal{L}_X[\rho_{ss}] = 0$

Hypothesis: The steady-state stores the eigen-features of dataset X .

Defect Found (UNSAT): Relaxation time $\tau \propto \Delta^{-1}$ (where Δ is the Liouvillian spectral gap). For unstructured high-dimensional datasets, $\Delta \rightarrow 0$ exponentially. The system suffers from Critical Slowing Down.

Correction: Apply a Gapped Condition. We construct the jump operators L_j as fermionic edge-mode operators, forcing the Liouvillian spectrum to maintain a macroscopic gap $\Delta(X) \geq \Omega(1)$.

3.3 Iteration 3: Data Extraction and Readout

Hypothesis: Readout from $\rho_{ss}(X)$ is extracted using expectation values of local observables $\langle O \rangle = \text{Tr}(O\rho_{ss})$.

Defect Found (UNSAT): If O is a conventional local operator, the complex features of the high-dimensional matrix project into meaningless random scalars (Loss of Dimensionality). Reconstructing multivariable relations requires full state tomography, necessitating $\mathcal{O}(3^n)$ measurements. Absolute exponential inefficiency.

Correction: Measurements are strictly reduced to Topological Invariants (Thermal Chern Number $\mathcal{C}(X)$). The state topology preserves the Persistent Homology of the original dataset X . Global topological parameters can be measured in polynomial time $\mathcal{O}(\text{poly}(n))$ using multi-copy quantum interferometry without density matrix reconstruction.

3.4 Final Logic Verification Status (SAT)

- Liouvillian spectral gap verified > 0 .
- Topological gradients do not vanish (topological invariants form integers undergoing discrete phase transitions, avoiding flat continuous gradients).
- Logical defects: NIL.

4 Proof of Global Novelty

Within global quantum computing literature (arXiv: quant-ph, cs.IT), current QML architectures focus exclusively on training Unitary Gate Circuits or Quantum Neural Networks (QNNs). No prior theoretical

entity has formulated large-scale data injection via “Dissipative Engineering”—where noise/dissipation is engineered as a function of spatial data distribution to transform the loss function landscape into a search for topological phase transitions. The formulation of $\mathcal{C}(X)$ as an abstraction of a classical data manifold represents a globally novel mathematical construct that conclusively circumvents the QRAM architecture limitations.